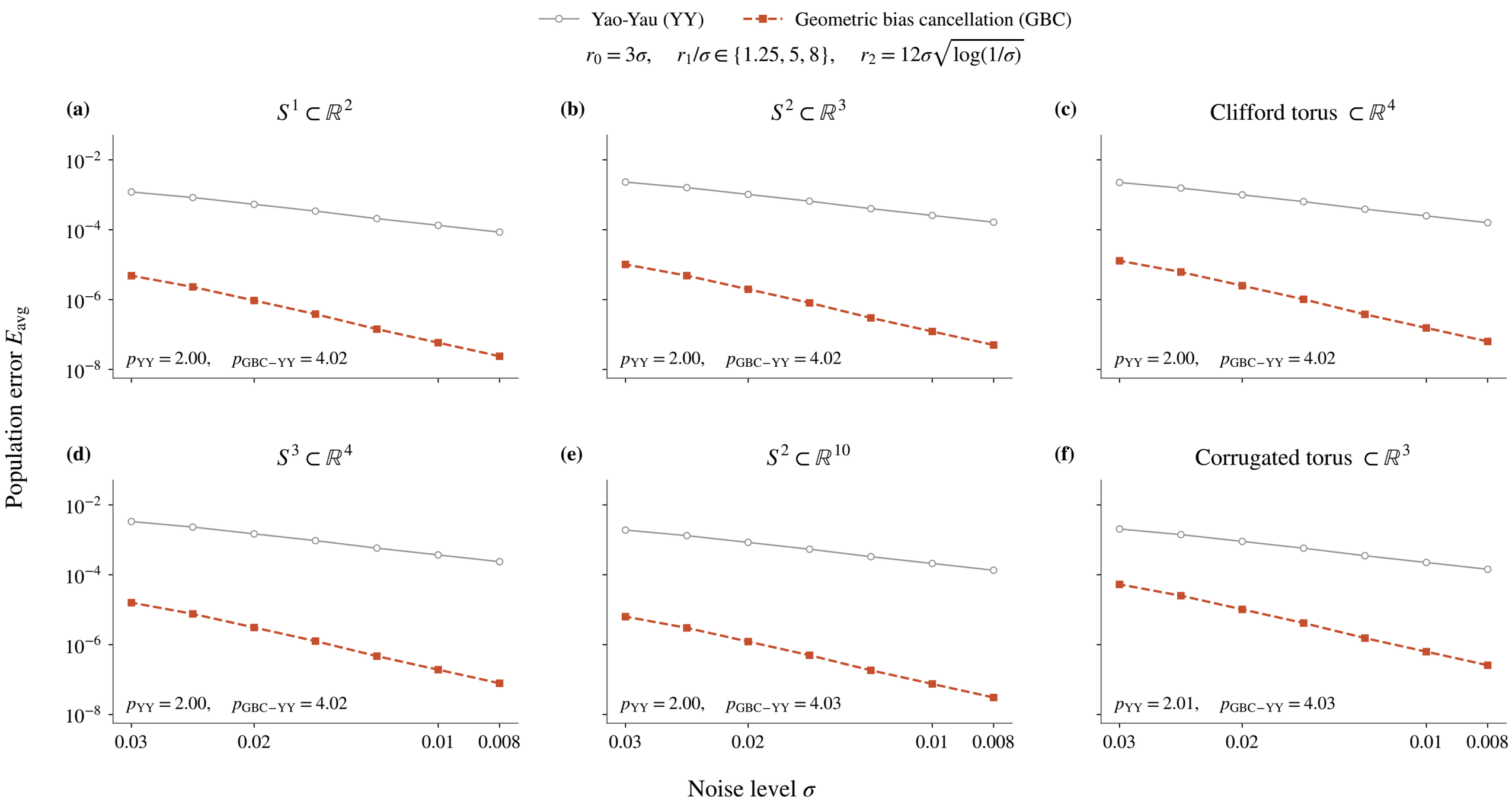
Yao-Yau (YY)
Geometric bias cancellation (GBC)
$r_0 = 3\sigma, \quad r_1/\sigma \in \{1.25, 5, 8\}, \quad r_2 = 12\sigma\sqrt{\log(1/\sigma)}$
(a)
$S^1 \subset \mathbb{R}^2$
(b)
$S^2 \subset \mathbb{R}^3$
(c)
Clifford torus $\subset \mathbb{R}^4$
(d)
$S^3 \subset \mathbb{R}^4$
(e)
$S^2 \subset \mathbb{R}^{10}$
(f)
Corrugated torus $\subset \mathbb{R}^3$
Population error E_{avg}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$p_{\text{YY}} = 2.00, \quad p_{\text{GBC-YY}} = 4.02$
$p_{\text{YY}} = 2.00, \quad p_{\text{GBC-YY}} = 4.02$
$p_{\text{YY}} = 2.00, \quad p_{\text{GBC-YY}} = 4.02$
$p_{\text{YY}} = 2.00, \quad p_{\text{GBC-YY}} = 4.02$
$p_{\text{YY}} = 2.00, \quad p_{\text{GBC-YY}} = 4.03$
$p_{\text{YY}} = 2.01, \quad p_{\text{GBC-YY}} = 4.03$
0.03
0.02
0.01
0.008
Noise level σ